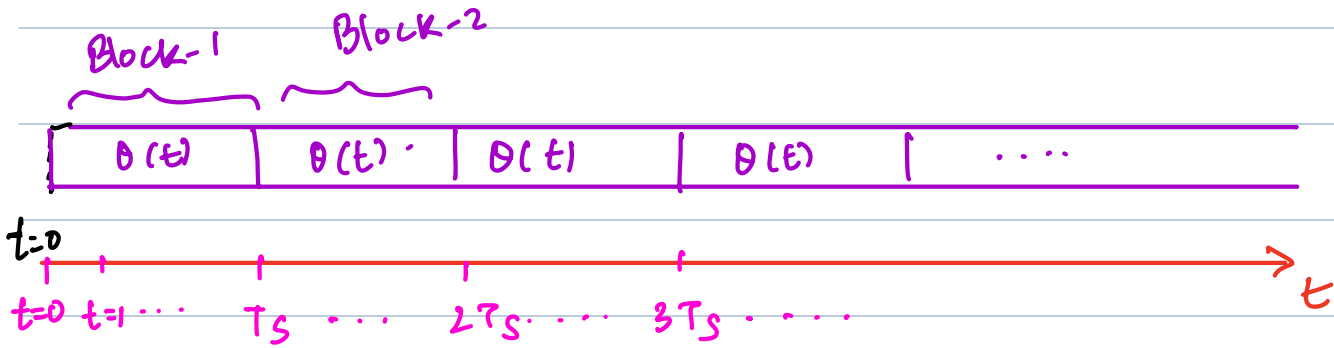


System & Scheme description:-



(*) Scheduling instants:- $t = 0, 1, 2, 3, \dots$ [slot-level basis]

(*) RIS switching instants:- $t = 0, T_s, 2T_s, 3T_s, \dots$ [Block-level]

1 block = T_s time slots. $T_s < \infty$

(*) Channel states:- $\{\underline{h}_1, \underline{h}_2, \dots, \underline{h}_M\}$; $\underline{h}_m \in \mathbb{C}^N$ is the cascaded fading channel state via N -element RIS.

Finite channel state space:-

→ Analysis easier

→ Justified under LOS propagation

(*) N channel paths are resolvable! "Rayleigh resolution limit"

(*) Group UEs belonging to a resolvable bin [with all UEs in a bin having same channel.]



(*) RIS states:- $\{\theta_1, \theta_2, \dots, \theta_L\}$. Q: PMF? $\theta_1, \dots, \theta_L$? $p_1, \dots, p_L$? L ?

Theorem 1: The probability density function for drawing the random samples of the IRS vector in every time slot to ensure that a PF scheduler achieves the BF-SE as in Lemma 2 is

$$f_{\theta}^{\text{opt}}(\theta') = \int_{\mathcal{F}^{-1}(\theta')} \delta(\mathbf{h} - \mathcal{F}^{-1}(\theta')) p_{\mathbf{h}_f}(\mathbf{h}) d\mathbf{h},$$

where $\mathcal{F}^{-1}(\theta')$ denotes the set-inverse under the $\mathcal{F}(\cdot)$ mapping, i.e., $\mathcal{F}^{-1}(\theta') \triangleq \{\mathbf{h} \in \mathbb{C}^{N+1} : \mathcal{F}(\mathbf{h}) = \theta'\}$, and $p_{\mathbf{h}_f}(\mathbf{h})$ is the probability density function of $\mathbf{h}_f = [h_d, (\mathbf{h}_1 \odot \mathbf{h}_2)^T]^T$ with $h_d \sim \mathcal{CN}(0, 1)$, $\mathbf{h}_1$ as given in (1), and $\mathbf{h}_2 \sim \mathcal{CN}(\mathbf{0}, \Sigma)$.

Ref: L. Yashvanth, C.R. Murthy, B.D. Rao, IEEE Wireless Comm. Letters, Oct. 2025.

↳ Takeaway:- The PMF of RIS states matches the PMF of channel states under $f(\cdot)$ mapping.

(*) Total of K UEs.

Probability that UE- k assumes channel state m :-

$$\Pr \{ \underline{h}_k' = \underline{h}_m \} = p_m, \quad m = 1, 2, \dots, M$$

[diff. for diff UEs!]

(*) PMF of RIS:- ($L = M$).

$$\underline{\Theta} = \{ \underline{\Theta}_1, \underline{\Theta}_2, \dots, \underline{\Theta}_M \}, \text{ where}$$

$\underline{\Theta}_m$ is chosen as the BF configuration vector for state m :-

$$\underline{\Theta}_m = \arg \max_{\underline{\Theta} \in \mathcal{G}} \{ \underline{\Theta}^H \underline{h}_m \}^2$$

$\underline{\Theta} \in \mathcal{G} \longrightarrow$ constraint set for RIS config.

and $\Pr \{ \underline{\Theta}(t) = \underline{\Theta}_m \} = p_m \quad \forall t = 0, T_s, \dots$
 $\forall m = 1, 2, \dots, M.$

Performance Analysis:-

(*) $\psi_m^{(K)} \rightarrow$ Fraction of K UEs whose channel state is $\underline{h}_m$.

$$(i.e) \quad \psi_m^{(K)} = \frac{1}{K} \sum_{k=1}^K \mathbb{1}_{\{ \underline{h}_k' = \underline{h}_m \}}.$$

Observation:- $\lim_{K \rightarrow \infty} \psi_m^{(K)} = \lim_{K \rightarrow \infty} \frac{1}{K} \sum_{k=1}^K \mathbb{1}_{\{ \underline{h}_k' = \underline{h}_m \}} \stackrel{LLN}{=} \mathbb{E} [\mathbb{1}_{\{ \underline{h}_k' = \underline{h}_m \}}] = \Pr \{ \underline{h}_k' = \underline{h}_m \} = p_m$

$$\Rightarrow \boxed{ \lim_{K \rightarrow \infty} \psi_m^{(K)} = p_m, \quad \forall m = 1, 2, \dots, M } \rightarrow \textcircled{1}$$

(*) Class- m : $k_m \triangleq \{ k \in \{1, 2, \dots, K\} : \underline{h}_k' = \underline{h}_m \}$. $\leftarrow$ set of UEs in channel state $\underline{h}_m$

$$\therefore |K_m| = \Psi_m^{(K)} K.$$

Beamforming (BF) Scheduling:-

Schedule a UE belonging to a class to which the RIS also points to. If there is more than UE, uniformly select one UE at random [anyways all UEs within a class will share the same channel quality.]

- e.g. do RR scheduling across UEs within a class!

Remark:- Under this scheduling scheme, the scheduler will continuously keep scheduling UEs from a single class over a T_s -time window.

Average Throughput of a UE-K in Class-m:-

$$T_{K,BF}^{(K)} = P_m \times \text{Avg. trpt over a window of } T_s$$

(in class m)

$$= P_m \times \text{Avg. Trpt of UE-K in } T_s \text{ time}$$

Prob. that RIS resides in class-m

$$= \frac{P_m}{T_s} \times E \left[\sum_{t=1}^{T_s} \text{Trpt at time } t \cdot \mathbb{1}_{\{K \text{ is scheduled UE at time } t\}} \right]$$

$$= \frac{P_m}{T_s} \times E \left[\sum_{t=1}^{N_K} R_m^{BF} \right]; \quad R_m^{BF} = \text{Rate under BF config.}$$

$N_K = \text{No. of slots scheduled for UE-K.}$

$$= \frac{P_m}{T_s} \cdot E[N_K] \cdot R_m^{BF}.$$

Note:- N_k is a random variable. & N_k 's are i.i.d. across $k \in [K_m]$.

But $\sum_{k \in K_m} N_k = T_s$; Taking expectations on both sides.

$$|K_m| \mathbb{E}[N_k] = T_s \Rightarrow \mathbb{E}[N_k] = \frac{T_s}{|K_m|}$$

Alternatively, do RR across all UEs $\Rightarrow N_k = \frac{T_s}{|K_m|}$ [give equal time to all UEs in the class]

$$\therefore \text{Avg. Trpt of UE-}k = \frac{P_m}{T_s} \times \frac{T_s}{|K_m|} \times R_m^{BF} = \frac{P_m R_m^{BF}}{\psi_m^{(k)} k}$$

$$\therefore \boxed{\begin{matrix} (k) & BF \\ T_{k,BF} = \frac{P_m R_m}{\psi_m^{(k)} k} \end{matrix}} \rightarrow (2).$$

Proportional-fair scheduling:-

Here, at every t :- $k^*(t) = \arg \max_{k=1.., K}$

PF metric of UE- k .

$$\frac{R_k(t)}{T_k(t)}$$

$$R_k(t) = \log_2 \left(1 + |h_k(t)|^2 \frac{P}{\sigma^2} \right)$$

$$T_k(t+1) = \left(1 - \frac{1}{T_c} \right) T_k(t) + \frac{1}{T_c} R_k(t) \mathbb{1}_{\{k = k^*(t)\}}.$$

Let $\beta_m^{(k)}$ be the fraction of time invested by PF scheduler in scheduling a UE from class- m .

Observation:- $\beta_m^{(k)} \geq 0$ & $\sum_{m=1}^M \beta_m^{(k)} = 1$; (i.e) Sum of fractions of time spent in all states = 1.

$\therefore \beta \triangleq [\beta_1^{(k)}, \beta_2^{(k)}, \dots, \beta_M^{(k)}]^T$ is a valid PMF!

$\therefore$ Avg. Trpt seen by UE-K in class m:-

$$T_{R, PF}^{(K)} \text{ (in class m)} = \beta_m^{(K)} \cdot \frac{R_{K, m}}{U_m^{(K)} K}$$

Rate achieved by UE-K in class-m.

Fraction of time PF scheduler selects a UE from class-m

These many UEs are equally likely to be selected in class-m as they share the same PF metric

Now:-

$$T_{R, PF}^{(K)} \text{ (in class m)} \leq T_{R, PF}^{(K)*} \triangleq \frac{\beta_m^{(K)} R_m^{BF}}{U_m^{(K)} K} \rightarrow (3)$$

Equality if RIS process also resides in class-m when PF-scheduler schedules a UE from class-m.

Property of a PF scheduler:-

Let $T_K^{(K)}$ be the long-term Avg. Trpt of UE-K out of K UEs.

The PF-algorithm with $t_c = \infty$ maximizes

$$R^{(K)} \triangleq \sum_{K=1}^K \log T_K^{(K)} \text{ among all class of schedulers!} \rightarrow (4)$$

$\therefore$ When $t_c \rightarrow \infty$:-

$$\sum_{K=1}^K \log T_{K, BF}^{(K)} \leq \sum_{K=1}^K \log T_{K, PF}^{(K)} \leq \sum_{K=1}^K \log T_{K, PF}^{(K)*}$$

From (4) From (3) $\rightarrow (A)$

$$\Rightarrow \sum_{m=1} \sum_{k \in K_m} \log T_{k, BF}^{(k)} \leq \sum_{m=1} \sum_{k \in K_m} \log T_{k, PF}^{(k)*}$$

From (2), $T_{k, BF}^{(k)} = \frac{p_m R_m}{\psi_m^{(k)} k}$ & (3) $\Rightarrow T_{k, PF}^{(k)*} = \frac{\beta_m^{(k)} R_m}{\psi_m^{(k)} k}$

independent of k within a class!

$$\Rightarrow \sum_{m=1}^M |K_m| \log \left(\frac{p_m R_m}{\psi_m^{(k)} k} \right) \leq \sum_{m=1}^M |K_m| \log \left(\frac{\beta_m^{(k)} R_m}{\psi_m^{(k)} k} \right)$$

$$\Rightarrow \sum_{m=1}^M \psi_m^{(k)} \log \left(\frac{p_m R_m}{\psi_m^{(k)} k} \right) \leq \sum_{m=1}^M \psi_m^{(k)} \log \left(\frac{\beta_m^{(k)} R_m}{\psi_m^{(k)} k} \right)$$

$$\Rightarrow \sum_{m=1}^M \psi_m^{(k)} \log \left(\frac{p_m}{\beta_m^{(k)}} \right) \leq 0$$

Take $\lim_{k \rightarrow \infty}$; then from (1) $\psi_m^{(k)} \xrightarrow[k \rightarrow \infty]{} p_m$.

$$\Rightarrow \sum_{m=1}^M p_m \log \left(\frac{p_m}{\beta_m^{(k)}} \right) \leq 0 \quad \text{as } k \rightarrow \infty$$

KL divergence b/w $\underline{p} = [p_1, \dots, p_M]$ & $\underline{\beta}$.

KL divergence ≥ 0

$$\Rightarrow \sum_{m=1}^M p_m \log \frac{p_m}{\beta_m^{(k)}} = 0 \quad \text{as } k \rightarrow \infty$$

$$\Rightarrow \lim_{k \rightarrow \infty} \beta_m^{(k)} = p_m, \quad \forall m=1,2,\dots,M \rightarrow \textcircled{5}$$

→ With large UEs, the fraction of time spent by PF scheduler in class m

Important! =
fraction of time spent by RIS state process in class m

Put $\textcircled{5}$ in $\textcircled{A}$:-

$$\sum_{m=1}^M \psi_m^{(k)} / k \log \left(\frac{p_m R_m^{BF}}{\psi_m^{(k)} k} \right) \leq \sum_{m=1}^M \psi_m^{(k)} / k \log T_{k,PF}^{(k)} \leq \sum_{m=1}^M \psi_m^{(k)} / k \log \left(\frac{\beta_m R_m^{BF}}{\psi_m^{(k)} k} \right)$$

R_m is a typical UE in class- m

Taking $k \rightarrow \infty$ & from $\textcircled{1}$: $\lim_{k \rightarrow \infty} \psi_m^{(k)} = p_m$
throughout: from $\textcircled{5}$: $\lim_{k \rightarrow \infty} \beta_m^{(k)} = p_m$

$$\sum_{m=1}^M p_m \log \left(\frac{R_m^{BF}}{k} \right) \leq \sum_{m=1}^M p_m \log T_{k,PF}^{(k)} \leq \sum_{m=1}^M p_m \log \left(\frac{R_m^{BF}}{k} \right)$$

By Sandwich Theorem,

$$\lim_{k \rightarrow \infty} \left(T_{k,PF}^{(k)} \text{ (in class } m) - \frac{R_m^{BF}}{k} \right) = 0$$

∴ Summing over all VEs!

$$\lim_{K \rightarrow \infty} \left(\sum_{k=1}^K T_{R, PF}^{(k)} - \frac{1}{K} \sum_{k=1}^K R_k^{BF} \right) = 0$$

↑
Avg s/s Trpt
obtained under
random RIS +
PF scheduling with $t_c \rightarrow \infty$.

↓
Avg. s/s Trpt obtained under RR
Scheduling + optimized RIS

∴ The result holds for ANY $t_s < \infty$ as long as $t_c \rightarrow \infty$

Intuition: -

As $t_c \rightarrow \infty$, and we run-the scheduler for long term, eventually, for any finite T_s , we will let the RIS to witness every possible state & ∴ $t_c \rightarrow \infty$, the PF scheduler will exploit the opportunity to keep scheduling VEs of a class to which the RIS points!

$T_s = \infty$ is not allowed! Convergence breaks!

Qn:- Can we relax the requirement on " $K \rightarrow \infty$ " in the proof!

Ans:- Yes! The origin of $K \rightarrow \infty$ is to use

$$\lim_{K \rightarrow \infty} \psi_m^{(K)} = P_m; \text{ which is then caused}$$

$$\lim_{K \rightarrow \infty} \beta_m^{(K)} = P_m!$$

$\therefore$ Instead of trying to match PMF of RIS states to actual PMF of channel, match it to "empirical PMF of channels"

(i.e) Let $\Pr\{\theta(t) = \theta_m\} = \psi_m^{(K)}, m=1, \dots, M$ $\rightarrow$ (i.e) the fraction of VEs in class- m .

(Clearly $\psi_m^{(K)} \geq 0$ & $\sum_{m=1}^M \psi_m^{(K)} = 1$ [i.e) sum of fraction of VEs in all states = 1])

Design choice!

$K \rightarrow \infty$ is not needed

$\rightarrow$ This's exactly what we're doing in our experiments now!

Analysis with Finite T_c :-

Q:- How large a T_c is needed?

Two key ingredients are needed to satisfy in the above proof!

① The RIS vectors should have visited all the different states sufficient # of time for reliable value of Avg. Trpt seen by a UE in a given class! In fact, every UE's Trpt should've been averaged well!

Suppose that there a total of $T = T_c$ slots:-

$$T_c \geq \max \left\{ mK, C \log \left(\frac{M}{\epsilon} \right) \right\}$$

(*) Diff UEs @ diff. distances ?

(*) Common PMF across UEs for all states !